Utilizing Generative AI and LLM to Create Neurological Physical Therapy Recommendations

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Background

- In occupational therapy, there is a pressing demand for efficiency due to time restrictions in clinical visits by stroke outpatients.
- This research attempts to utilize the generative abilities of artificial intelligence to instantaneously generate a Home Exercise Program for the stroke outpatient and refers to the Rehabilitation Treatment Specification System (RTSS) to help determine which exercises are best for the stroke patient.
- The Home Exercise Program contains videos of home exercises tailored for the stroke outpatient individual to follow along with at home. All the videos are created by Hep2go Inc's YouTube channel. Hep2go is a clinician-recommended database of various exercises for physical therapy.

Cosine Similarity

Suppose: $a =< x_1, y_1 >$ and $b =< x_2, y_2 >$

The magnitudes for a, b, c...

$$|c| = \sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2}$$

$$|a| = \sqrt{(x_1^2 - 0) + (y_1^2 - 0)} = \sqrt{x_1^2 + y_1^2}$$

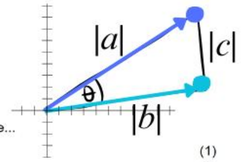
$$|b| = \sqrt{(x_2^2 - 0) + (y_2^2 - 0)} = \sqrt{x_2^2 + y_2^2}$$

Law of cosine

$$c^2 = a^2 + b^2 - 2abcos(\theta)$$

Dot product of a and b

$$a \cdot b = (x_1 x_2) + (y_1 y_2)$$



Plugging magnitudes of a, b, c into law of cosine...

$$|c|^2 = |a|^2 + |b|^2 - 2|a||b|cos(\theta) \quad (1)$$

$$2|a||b|cos(\theta) = |a|^2 + |b|^2 - |c|^2 \quad (2)$$

$$2|a||b|cos(\theta) = \left(\sqrt{x_1^2 + y_1^2}\right)^2 + \left(\sqrt{x_2^2 + y_2^2}\right)^2 - \left(\sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2}\right)^2 \quad (3)$$

$$2|a||b|cos(\theta) = \left(x_1^2 + y_1^2\right) + \left(x_2^2 + y_2^2\right) - \left[(x_1 - x_2)^2 + (y_1 - y_2)^2\right]$$

$$2|a||b|cos(\theta) = \left(x_1^2 + y_1^2\right) + \left(x_2^2 + y_2^2\right) - \left[\left(x_1^2 - 2x_1x_2 + x_2^2\right) + \left(y_i^2 - 2y_1y_2 + y_j^2\right)\right] \quad (4)$$

$$2|a||b|cos(\theta) = \left(x_1^2 - x_1^2 + y_1^2 - y_i^2\right) + \left(x_2^2 - x_2^2 + y_2^2 - y_2^2\right) - [(-2x_1x_2) + (-2y_1y_2)] \quad (5)$$

$$2|a||b|cos(\theta) = 2x_1x_2 + 2y_1y_2$$

$$cos(\theta) = \frac{2x_1x_2 + 2y_1y_2}{2|a||b|} = \frac{x_1x_2 + y_1y_2}{|a||b|} \quad (6)$$

Rewrite $x_1x_2 + y_1y_2$ as the dot product of a and b

$$\cos(\theta) = \frac{\mathbf{a} \cdot \mathbf{b}}{|\mathbf{a}||\mathbf{b}|} \quad (7)$$

Objectives

Institute a seamless pipeline for occupational therapists to efficiently recommend a Home Exercise Program to their stroke outpatient. The pipeline should intake an outpatient's conditions using Electronic Health records (EHR), then output a Home Exercise Program by using the text-generation abilities of gpt-3.5-turbo-0125, a Large Language Model (LLM) by OpenAI.

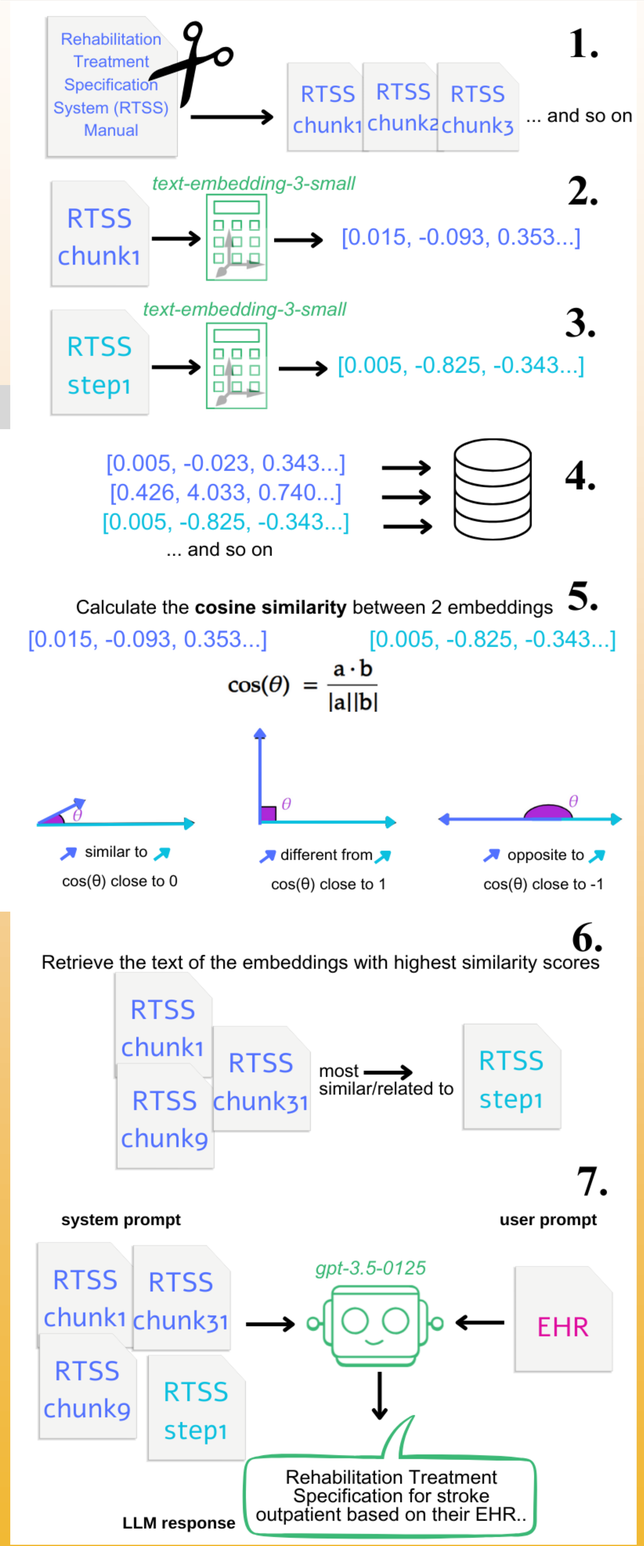
References

Esteva, M., & Bi, H. (2009). Inferring intra-organizational collaboration from cosine similarity distributions in text documents. In Proceedings of the 9th ACM/IEEE-CS joint conference on Digital libraries. JCDL '09: Joint Conference on Digital Libraries. ACM. https://doi.org/10.1145/1555400.1555475

Lewis, P. et al. "Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks," *Proceedings of the 34th International Conference on Neural Information Processing Systems, December 2020: 9459-9474.*

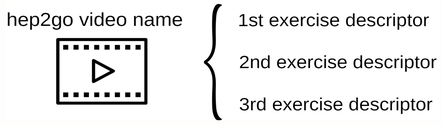
Van Stan, J. H. et al. "The Rehabilitation Treatment Specification System: Implications for Improvements in Research Design, Reporting, Replication, and Synthesis," *Archives of Physical Medicine and Rehabilitation*, vol. 100, pp. 146–155, 2019.

Fig 1, Methodology: Retrieval Augmented Generation (RAG) steps



Methodology (continued): Outputting videos

8. Web scrape to get all video links and titles from Hep2go Inc.'s YouTube channel.
9. Create common exercise descriptors based on exercise categories from Hep2go's website of exercises.
   - Stretch exercises
   - Active Range of Motion (AROM)
   - Elastic band exercises
   - etc.



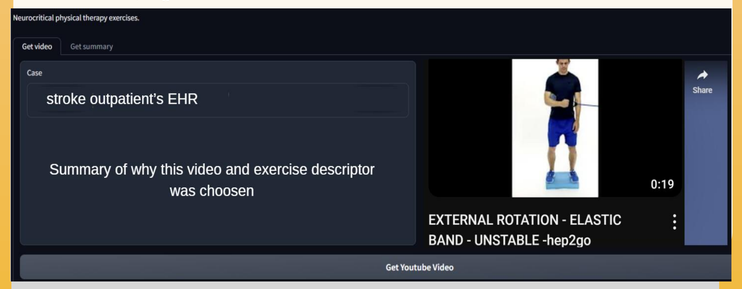
10. Repeat steps 2. through 6 (refer to fig 1.), replacing:
    - "RTSS chunk" and its embeddings with Hep2go video titles and its embeddings.
    - "RTSS step" and its embeddings with the common exercise descriptors and its embeddings.
11. Store each video title and link with its top 3 most similar/related to exercise descriptors
12. Create a prompt asking LLM to select the 3 exercise descriptors that are most optimal for the stroke outpatient given their rehabilitation treatment specification and EHR.
13. Output the hep2go video titles & links that contain any of the 3 exercise descriptors listed by the LLM's answer.

Output:

	Youtube title	Link	Exercise descriptor
2	Lat Stretch -hep2go	https://www.youtube.com/watch?v=hSCCfmdfOv8	Stretch exercises
3	STANDING CALF STRETCH - SOLEUS -hep2go	https://www.youtube.com/watch?v=-h-SqMK0KZo	Stretch exercises
4	WRIST RADIAL DEVIATION - AROM -hep2go	https://www.youtube.com/watch?v=tHldOfQz4yY	Active Range of Motion
5	WRIST RADIAL DEVIATION - AROM -hep2go	https://www.youtube.com/watch?v=cpiM78CSGME	Active Range of Motion
6	STANDING CALF STRETCH - GASTROC -hep2go	https://www.youtube.com/watch?v=Ekq9vHPwDaA	Stretch exercises
7	QUAD STRETCH - STANDING -hep2go	https://www.youtube.com/watch?v=KbVrqOJ-7Dg	Stretch exercises
8	HAMSTRING STRETCH WITH TOWEL -hep2go	https://www.youtube.com/watch?v=sRiM1X1cF_I	Stretch exercises
9	ELASTIC BAND - KNEE EXTENSIONS -hep2go	https://www.youtube.com/watch?v=JmGYCt8Ao0c	Stretch exercises
10	ELASTIC BAND - KNEE EXTENSIONS -hep2go	https://www.youtube.com/watch?v=t3FfZABq1Gw	Stretch exercises

Discussion

- Generating a more "personalized" video selection
- implement a user-friendly interface that displays the exercise video and a conversational explanation of why the hep2go exercise video was chosen as the output.



Acknowledgements

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